

Q. For what value of n , the matrix

$$\begin{bmatrix} 5-n & n+1 \\ 2 & 4 \end{bmatrix} \text{ is singular.}$$

$$|A| = ad - bc$$

$$(5-n)4 - (2)(n+1)$$

$$20 - 4n - 2n - 2$$

$$-6n + 18 = 0$$

$$-6n = -18$$

$$\boxed{n=3} \checkmark$$